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Pembrolizumab versus Ipilimumab in Advanced Melanoma (KEYNOTE-006)

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Background

Metastatic melanoma has historically been associated with a poor prognosis. The advent of immune checkpoint inhibitors has revolutionized treatment. Pembrolizumab (Keytruda), an anti-PD-1 antibody, has shown promising activity in early-phase studies. KEYNOTE-006 was designed to compare pembrolizumab with ipilimumab, the previous standard of care.

Study Design

In this open-label, multicenter, randomized, phase 3 trial, 834 patients with advanced melanoma were assigned to receive pembrolizumab (at a dose of 10 mg per kilogram of body weight) every 2 weeks or every 3 weeks or four doses of ipilimumab (3 mg per kilogram) every 3 weeks. The primary end points were progression-free survival and overall survival.

Clinical Outcomes

The estimated 6-month progression-free-survival rates were 47.3% for pembrolizumab every 2 weeks, 46.4% for pembrolizumab every 3 weeks, and 26.5% for ipilimumab (hazard ratio for disease progression, 0.58; $P < 0.001$). The estimated 12-month survival rates were 74.1%, 68.4%, and 58.2%, respectively (hazard ratio for death for pembrolizumab every 2 weeks vs. ipilimumab, 0.63; $P = 0.0005$; and for pembrolizumab every 3 weeks vs. ipilimumab, 0.69; $P = 0.0036$).

Discussion

Pembrolizumab demonstrated a significant survival advantage over ipilimumab. Response rates were significantly higher with pembrolizumab (33.7% and 32.9%) than with ipilimumab (11.9%). Benefit was observed across all subgroups, including those with BRAF mutations. These results emphasize the role of PD-1 blockade as a first-line treatment strategy for melanoma patients like P002, P011, and P017.

Conclusion

Pembrolizumab prolonged progression-free survival and overall survival and had less high-grade toxicity than did ipilimumab in patients with advanced melanoma. This is the evidence supporting the therapy for our Command Center's oncology cases.